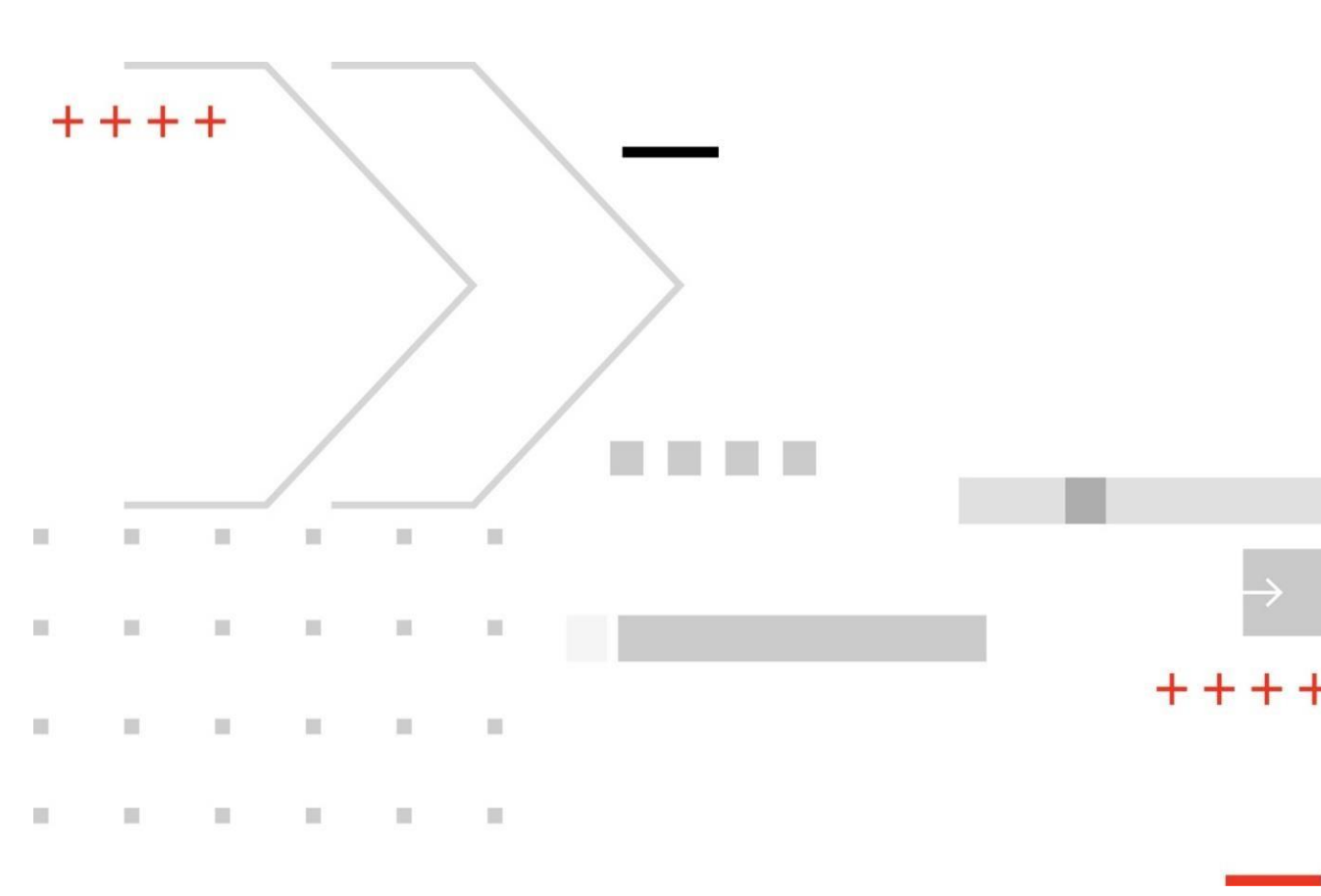


IN-LIFE OPERATIONS

VEEAM BACKUP CONFIGURATION FOR IMMUTABLE REPOSITORY

SYNOPSIS: This document describes the procedure performed to configure an immutable repository in Veeam Backup hosted on an HPE Alletra 4120 server with Debian 11 operating system.
REFERENCE: 100348 – Installation and configuration manual for services

VERSION: V1.2
DATE: October 2025
STATUS: In Progress



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1. INTRODUCTION

1.1 PURPOSE

Within the framework of infrastructure operations of the ICT CALA L1 team, the procedure for configuring an immutable repository in Veeram Backup & Replication, hosted on an HPE Alletra 4120 server with Debian 11 operating system, is described.

1.2 SCOPE

Secure the backup copies stored on the server without allowing modification or deletion during active periods of operation, thus strengthening protection against possible ransomware attacks or accidental data deletion.

1.3 INTENDED AUDIENCE

This document is intended for use in services provided to MS Claro Colombia by the Product Support (L1) team, Managed Services team, and L2 Support team.

1.4 ESSENTIAL SERVER CONCEPTS

- **Backup repository:** The physical or logical location where backups generated by Veeam are stored.
- **Immutability:** A property that prevents the modification or deletion of files for a defined period, even by users with administrative privileges.
- **Veeam Transport Service:** The component that manages data transfer between the backup server and the repository.
- **SSH (Secure Shell):** The protocol that allows encrypted remote connections to the Linux server.

Taking the above into account, the Veeam Backup configuration is established for the immutability of the Alletra 4120 server, complying with the implementation of services in virtualization environments.

2. Hardware and Software Requirements

Hardware Requirements (Alletra 4120 Server)

Server: HPE Alletra 4120, with storage controller, configured disks (RAID)

Storage: Disk space based on inventory availability (multiple disks, boot from RAID controller). In this case, RAID arrays:

- RAID 6 + 0 (8 disks – Array A)
- RAID 5 (8 disks – Array B)

Network connectivity: Network stability at the time of Veeam Backup configuration, direct connection via Ethernet port

Software Requirements

Operating system: Debian 11

Veeam Backup: Replication v12 or higher

3. TOPOLOGÍA DE RED

The implemented architecture is based on a remote server-repository model, where the main Veeam server (running in a Windows environment) communicates with the Debian server via SSH for backup transfer and management.

The Debian server acts as an immutable repository, with its respective RAID storage configured to hold the backups. Immutability retention policies (2-7 days for testing, or for months or years, as needed) are applied to the designated folder.

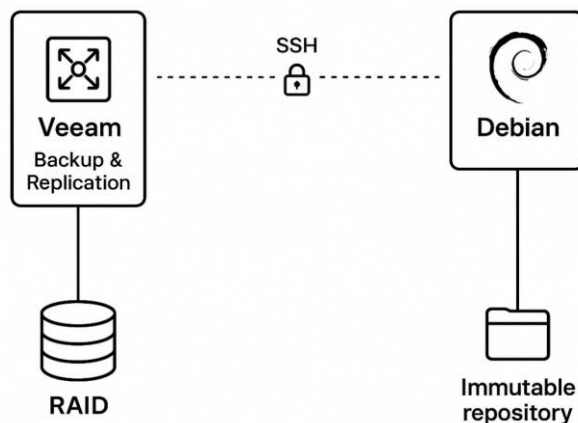


Figure 1. Network Topology

4. INSTALLATION AND CONFIGURATION OF THE SERVICE

4.1 Debian Server Configuration

1. Before performing any configuration, it is important to verify that the system correctly recognizes the previously created disks and RAID arrays. To do this, we use the `lsblk` command in the console.

This command displays the storage devices and their partitions, organized hierarchically.

In this case, three main storage units were identified on the server: /dev/sda

- /dev/sdb
- /dev/nvme0n1

These are associated with the disk partitions on the server.

```

root@debianAlletra:/home/veeamrepo# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINT
sda                                 8:0      0  36.4T  0 disk
├─sda1                             8:1      0   512M  0 part
├─sda2                             8:2      0   488M  0 part
├─sda3                             8:3      0   36.4T  0 part
│   ├─debian--vg-root              254:3      0   27.9G  0 lvm
│   ├─debian--vg-swap_1            254:4      0    976M  0 lvm  [SWAP]
│   └─debian--vg-home              254:5      0   36.4T  0 lvm
sdb                                 8:16     0  36.4T  0 disk
nvme0n1                            259:1     0  447.1G  0 disk
├─nvme0n1p1                       259:2     0    512M  0 part  /boot/efi
├─nvme0n1p2                       259:3     0    488M  0 part  /boot
├─nvme0n1p3                       259:4     0   446.1G  0 part
│   ├─debianAlletra--vg-root       254:0      0   27.9G  0 lvm  /
│   ├─debianAlletra--vg-swap_1     254:1      0    976M  0 lvm  [SWAP]
│   └─debianAlletra--vg-home       254:2      0   417.2G  0 lvm  /home
root@debianAlletra:/home/veeamrepo#
  
```

Figure 2. Partitions created

2. Installing parted, a utility that allows you to manage disk partitions on Linux/Unix systems. With parted you can create, delete, resize, or move partitions, as well as reorganize disk usage and copy data.

The following commands are executed:

```

$ sudo apt install parted
$ sudo parted -l
  
```

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```
root@debianAlletra:/home/veeamrepo# apt update
Hit:1 http://deb.debian.org/debian bullseye InRelease
Get:2 http://deb.debian.org/debian bullseye-updates InRelease [44.0 kB]
Get:3 http://security.debian.org/debian-security bullseye-security InRelease [27.2 kB]
Get:4 http://security.debian.org/debian-security bullseye-security/main Sources [250 kB]
Get:5 http://security.debian.org/debian-security bullseye-security/main amd64 Packages [389 kB]
Get:6 http://security.debian.org/debian-security bullseye-security/main Translation-en [259 kB]
Fetched 969 kB in 0s (2,158 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@debianAlletra:/home/veeamrepo# apt install parted
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libparted2
Suggested packages:
  libparted-dev libparted-i18n parted-doc
The following NEW packages will be installed:
  libparted2 parted
0 upgraded, 2 newly installed, 0 to remove and 70 not upgraded.
Need to get 554 kB of archives.
After this operation, 935 kB of additional disk space will be used.
Do you want to continue? [y/n] y
Get:1 http://deb.debian.org/debian bullseye/main amd64 libparted2 amd64 3.4-1 [335 kB]
Get:2 http://deb.debian.org/debian bullseye/main amd64 parted amd64 3.4-1 [219 kB]
Fetched 554 kB in 0s (1,914 kB/s)
Selecting previously unselected package libparted2:amd64.
(Reading database ... 27935 files and directories currently installed.)
Preparing to unpack .../libparted2_3.4-1_amd64.deb ...
Unpacking libparted2:amd64 (3.4-1) ...
Selecting previously unselected package parted.
Preparing to unpack .../parted_3.4-1_amd64.deb ...
Unpacking parted (3.4-1) ...
Setting up libparted2:amd64 (3.4-1) ...
Setting up parted (3.4-1) ...
Processing triggers for libc-bin (2.31-13+deb11u10) ...
Processing triggers for man-db (2.9.4-2) ...
root@debianAlletra:/home/veeamrepo# parted /dev/sdb
GNU Parted 3.4
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mklabel gpt
```

Figure 3. Installation of "parted"

3. Detect unpartitioned disks and create a partition table (if a disk was missing). Additionally, identify if any disk still lacks a partition table; if so, use...

```
$ sudo fdisk /dev/sdx
```

This will display all detected disks and their partitions. If a disk appears with the message "Partition Table unknown", a new partition table needs to be created for it.

4. Open parted on the corresponding disk, in this case /dev/sdb, the partition table will be created in GPT format and then a partition that uses the entire disk.

5. The partitions are created by entering: *parted /dev/sdb*

```
root@debianAlletra:/home/veeamrepo# parted /dev/sdb
GNU Parted 3.4
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
```

Figure 4. Parted

6. Create a GPT partition table to erase everything you can see and leave the disk ready to use GPT, with the command: *mklabel gpt*

This command is essential for preparing a disk for use, replacing any existing partition table and establishing the new format. It is especially useful for disks larger than 2 TB or modern systems with UEFI, as GPT is the most current standard for disk partitioning.

7. Create a primary partition that uses the entire disk. With the same parted environment open, create a primary partition formatted as ext4, using all available space.

```
$mkpart primery ext4 0% 100%
```

mkpart – Create a new partition

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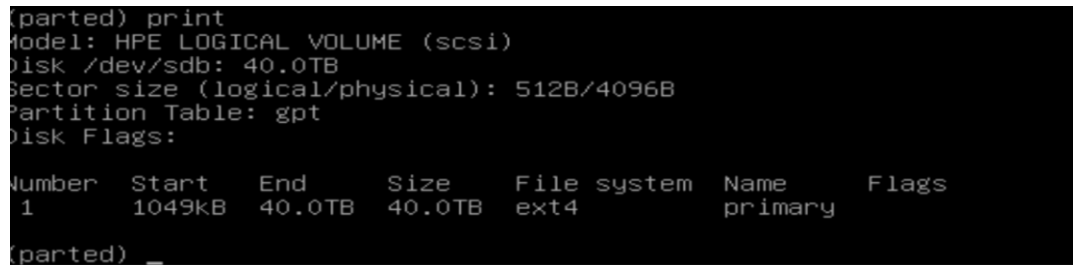
primary – It indicates that it will be a primary partition.

ext4 – Define the standard file system, designed specifically for the Linux permissions and security structure

mkpart creates a partition of a specified type, in this case primary, which is used to define the partition table structure. This command is part of the 'parted' utility in Linux. This type of partition is fundamental to operating systems and can contain system files.

8. Verify that the partition was created correctly by running: `print` .

This will display disk information, partition table, and allocated size. You should see something similar to the following. To exit parted, type the command *quit*.



```
(parted) print
Model: HPE LOGICAL VOLUME (scsi)
Disk /dev/sdb: 40.0TB
Sector size (logical/physical): 512B/4096B
Partition Table: gpt
Disk Flags:

Number   Start    End      Size    File system  Name     Flags
  1       1049kB   40.0TB   40.0TB   ext4         primary

(parted) _
```

Figure 5. Print

9. Format the new partition. Once the partition is created, it must be formatted with the ext4 file system. This option erases all data on that partition. In this case, there is no risk since the disk has just been created.

The following command will be used to format the mkfs.ext extension: `/dev/sdb1`

10. Create a mount point (for Veeam backups): Create a folder where the new disk will be mounted. In this case, it will be used for Veeam backups.

```
$ mkdir /home/veeamrepo
```

* The name or path can be modified as needed, for example `/Backup` or `/home/Backup_veeam`.

11. Mount the partition manually to verify that it is working correctly:

```
$ mount /dev/sdb1 /home/veeamrepo
```

12. Mount the partition manually to verify that it is working correctly:

```
$ mount /dev/sdb1 /home/veeamrepo
```

To verify the assembly, we validate the disk information: `sdf -h`

It should show you the partition mounted at the path: `/home/veeamrepo`

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```
root@debianAlletra:/home/veeamrepo# df -h
Filesystem                Size      Used Avail Use% Mounted on
udev                     32G         0   32G   0% /dev
tmpfs                     6.3G      1.3M   6.3G   1% /run
/dev/mapper/debianAlletra--vg-root 28G      1.3G    25G   5% /
tmpfs                     32G         0   32G   0% /dev/shm
tmpfs                     5.0M         0   5.0M   0% /run/lock
/dev/mapper/debianAlletra--vg-home 410G       92K   389G   1% /home
/dev/nvme0n1p2            471M       50M   397M  12% /boot
/dev/nvme0n1p1            511M       5.8M   506M   2% /boot/efi
tmpfs                     6.3G         0   6.3G   0% /run/user/0
root@debianAlletra:/home/veeamrepo#
```

Figure 6. Disk partitions

13. To configure an automatic message at system startup, so that the disk is automatically mounted when the server starts, you must edit the configuration file: **/etc/fstab**

The following command lines are added:

```
GNU nano 5.4 /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
#<file system> <mount point> <type> <options> <dump> <pass>
dev/mapper/debianAlletra--vg-root / ext4 errors=remount-ro 0 1
/boot was on /dev/nvme0n1p2 during installation
UUID=cc402ca6-9dac-4c8b-9c00-fada602586de /boot ext2 defaults 0 2
/boot/efi was on /dev/nvme0n1p1 during installation
UUID=8CAF-934D /boot/efi vfat umask=0077 0 1
dev/mapper/debianAlletra--vg-home /home ext4 defaults 0 0 2
dev/mapper/debian--vg-swap_1 none swap sw 0 0 0
dev/mapper/debianAlletra--vg-swap_1 none swap sw 0 0 0
dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
```

Figure 7. fstab configuration

Within this file, each line represents a partition and uses six columns to configure it:

Device (UUID/Nombre): Identify the physical partition.

Mount Point: The folder where the files will appear.

File System Type: The partition format (ext4, ntfs, swap).

Options: Assembly rules (e.g. Defaults, ro for read only, noatime, etc.).

Dump: Used by backup tools (usually 0).

Pass: Used by backup tools.

4.1 SSH configuration for connecting to Veeam

1. The following commands are executed on the main server:

1.1 Verify that the SSH service is active:

```
$ sudo systemctl status ssh
```

1.2 If the SSH service is not running, run the following:

```
$ sudo systemctl enable --now ssh
```

2. Test connection from the Veeam server:

\$ ssh user_name@ip_server

3. If there were errors, the permissions in /etc/ssh/sshd_config and the firewall rules were checked.

4.2 Veeam Immutable Repository Configuration

1. Log in to Veeam: Open the Veeam Backup & Replication console and log in with a user with administrative permissions.

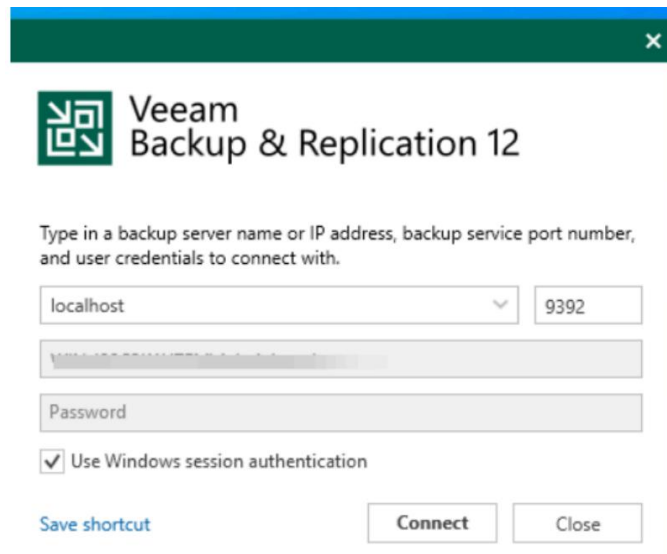


Figura 8. Inicio de Sesión Veeam

2. Add a new Linux Repository. From the main console, go to:

Backup Infrastructure > Backup Repositories > Add Repository

- Select the option Direct attached storage -> Linux Repository
- Enter a name for the repository
3. Authenticate using the Debian server's SSH credentials
- Veeam will connect to the server via SSH to verify connectivity and permissions
- Verify that the SSH service is active and that the user has access to the path where the backups will be stored.
4. Specify the storage path in the Veeam wizard, indicating the folder mounted on the Linux server where the backups will be saved (/home/veeamrepo/Prueba). This path must have the appropriate permissions and sufficient space to accommodate the backup files.
5. Immutability Configuration:
In the Mount Server section, leave the Debian server as the default.
En repository > Advanced, check the box:

Make recent backups immutable for [X] days.

This value defines how many days the files will be immutable (unmodifiable and undeletable). For

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testing purposes, you can set it to a low number (between 1 and 2 days). In the same wizard, enable the "Single-use credentials for hardened repository" option.

This feature prevents SSH access credentials from being stored in the Veeam system, increasing repository security.

5. Create a Backup Job

Go to Home > Jobs > Backup Job > Virtual Machine (or select File or Linux computer, depending on the type of backup you want to test) In this case, a previously created virtual test machine will be used.

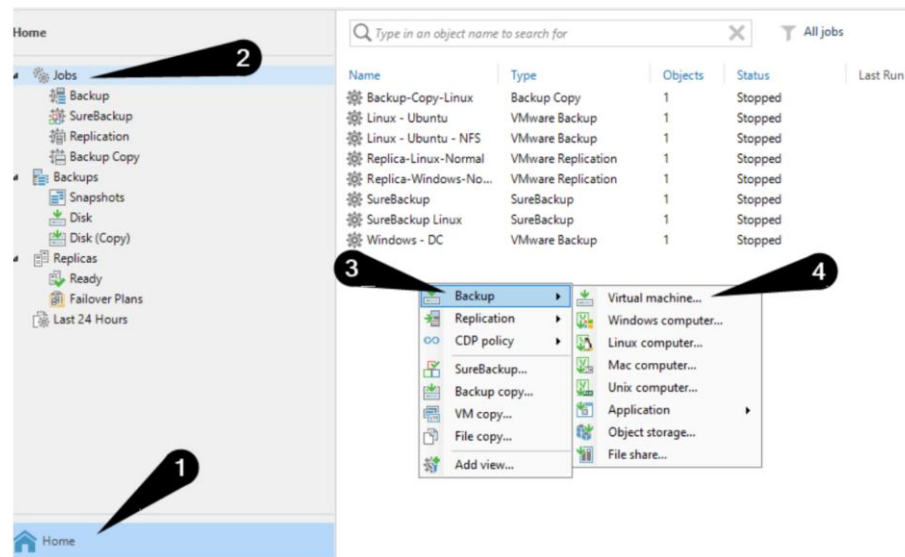


Figure 9. Job creation

Assign a name to the job, for example: PruebaInmutabilidad

6. In the same window, select the resources to back up (virtual machines, folders, or files).
7. Choose the configuration repository mentioned above as the location where the backup should be saved.

Define the restore point retention policy according to your needs. For testing purposes, the backup period can be set to the shortest possible.

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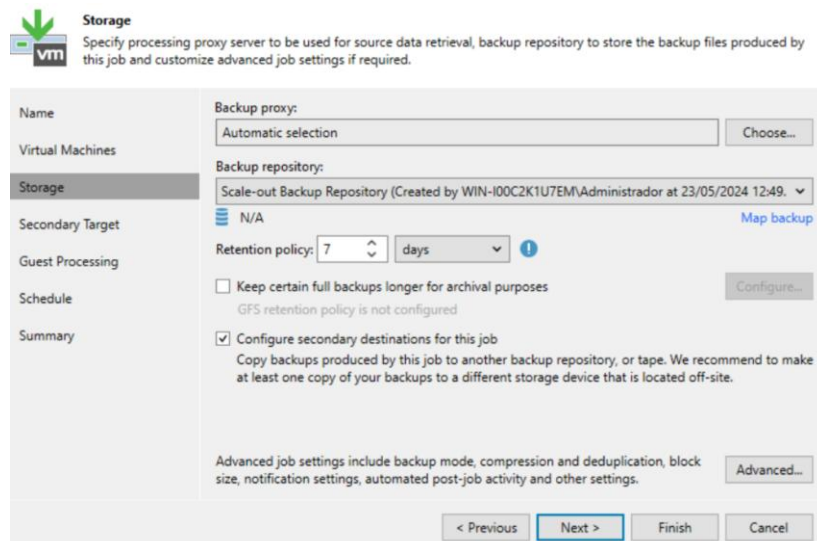


Figure 10. Job Parameters

9. Schedule the job to run immediately and verify its execution.
10. Review and finalize the process in Veeam.

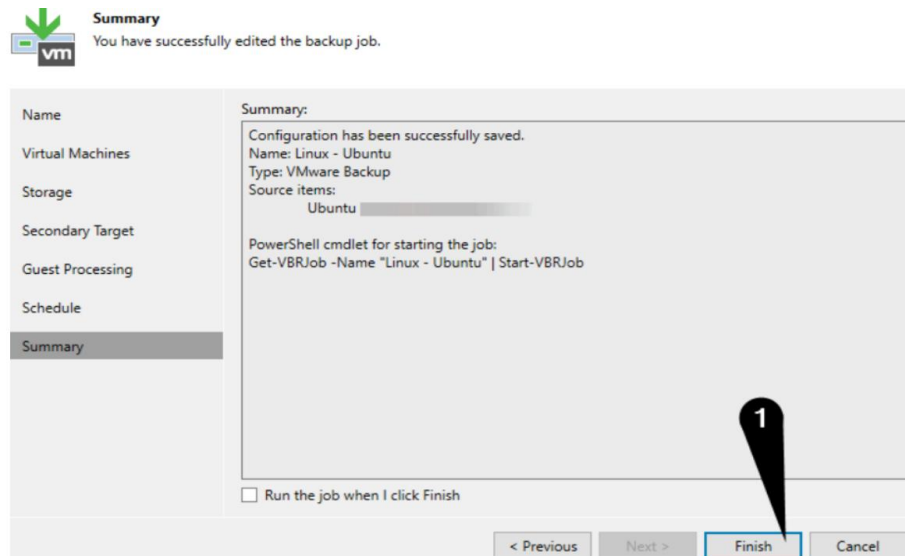


Figure 11. Finish Job Creation

When configuring the repository, in the Veeam wizard, select Single-use credentials. This prevents storing the credentials on the repository system.

11. Once the backup is complete, the server is checked to ensure that the backup files were created correctly and that they have the immutability attribute.

The following commands are executed for verification:

```
$ cd /home/veeamrepo/Prueba
```

This command allows you to access the folder where backups generated by Veeam are stored.

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To list all files in the directory, we use: `ls -la`

This command includes hidden files, showing their owner, permissions, and modification date -> `lsattr`

- If a file appears with an "i" in the attribute column, it means it is immutable.

- Immutable files cannot be modified, deleted, or renamed until the attribute is removed or the specified immutability period configured in Veeam is reached.

```
root@debianAlletra:~# cd /home/veeamrepo/Prueba/
root@debianAlletra:/home/veeamrepo/Prueba# ls -la
total 3581816
-rwxr-xr-x 2 veeamrepo veeamrepo 4096 Aug 6 17:19 .
-rwxr-xr-x 4 veeamrepo veeamrepo 4096 Aug 6 17:28 ..
-rw-r--r-- 1 veeamrepo veeamrepo 3667742720 Aug 6 17:19 monitoreo.62D2025-08-06T161310_5514.vbk
-rw-r--r-- 1 veeamrepo veeamrepo 16887 Aug 6 17:19 monitoreo_DB9CB.vbm
-rw-r--r-- 1 root root 176 Aug 6 17:19 .veeam.1.lock
root@debianAlletra:/home/veeamrepo/Prueba# lsattr
-----e----- ./monitoreo_DB9CB.vbm
---i-----e----- ./monitoreo.62D2025-08-06T161310_5514.vbk
```

Figure 12. Property of immutability

For this specific example, you can see the immutable file created "monitoreo.62D2025..." In the properties, you will see an "i" which means that it is an immutable file.

5. Test Plan

With the immutable repository configured, several tests were designed and executed to validate the proper protection of backups against unauthorized modifications, deletions, or alterations. These tests were initially performed as a user without administrative privileges and subsequently as the superuser (root) to determine the actual scope of the immutability established by Veeam.

5.1 Veeam Immutable Repository Configuration

Case 1: Attempt to modify files

An attempt was made to open and modify the contents of the file within the test file located in the folder configured as immutable.

Expected result: The system should deny write access and display an "Operation not permitted" message.

Actual result: The modification attempt was rejected, confirming that the files could not be altered.

```
root@debianAlletra:~# cd /home/veeamrepo/Prueba/
root@debianAlletra:/home/veeamrepo/Prueba# ls -la
total 3581816
-rwxr-xr-x 2 veeamrepo veeamrepo 4096 Aug 6 17:19 .
-rwxr-xr-x 4 veeamrepo veeamrepo 4096 Aug 6 17:28 ..
-rw-r--r-- 1 veeamrepo veeamrepo 3667742720 Aug 6 17:19 monitoreo.62D2025-08-06T161310_5514.vbk
-rw-r--r-- 1 veeamrepo veeamrepo 16887 Aug 6 17:19 monitoreo_DB9CB.vbm
-rw-r--r-- 1 root root 176 Aug 6 17:19 .veeam.1.lock
root@debianAlletra:/home/veeamrepo/Prueba# lsattr
-----e----- ./monitoreo_DB9CB.vbm
---i-----e----- ./monitoreo.62D2025-08-06T161310_5514.vbk
root@debianAlletra:/home/veeamrepo/Prueba# echo monitoreo.62D2025-08-06T161310_5514.vbk >> /home/veeamrepo/Prueba/monitoreo.62D2025-08-06T161310_5514.vbk
bash: /home/veeamrepo/Prueba/monitoreo.62D2025-08-06T161310_5514.vbk: Operation not permitted
root@debianAlletra:/home/veeamrepo/Prueba#
```

Figure 13. Modification test

Case 2: Attempt to modify files

An attempt was made to delete the previous file using the command: `rm monitoreo.62D2025...vbk`

Expected result: The system should prevent deletion due to accountability policies.

Actual result: The system denied the action, displaying “Operation not permitted,” confirming that the policy was active.

```
root@debianAlletra:/home/veeamrepo/Prueba# ls -la
total 3581816
-rwxr-xr-x 2 veeamrepo veeamrepo 4096 Aug 6 17:19 .
-rwxr-xr-x 4 veeamrepo veeamrepo 4096 Aug 6 17:28 ..
-rw-r--r-- 1 veeamrepo veeamrepo 3667742720 Aug 6 17:19 monitoreo.62D2025-08-06T161310_5514.vbk
-rw-r--r-- 1 veeamrepo veeamrepo 16887 Aug 6 17:19 monitoreo_DB9CB.vbm
-rw-r--r-- 1 root root 176 Aug 6 17:19 .veeam.1.lock
root@debianAlletra:/home/veeamrepo/Prueba# rm monitoreo.62D2025-08-06T161310_5514.vbk
rm: cannot remove 'monitoreo.62D2025-08-06T161310_5514.vbk': Operation not permitted
root@debianAlletra:/home/veeamrepo/Prueba#
```

Figure 14. Elimination test

Case 3: Attempt to move files

An attempt was made to move an immutable file to another location within or outside the repository: `mv monitoreo.62D2025...vbk`

Expected result: the file should remain in its original location without alteration.

Actual result: the system blocked the move operation, ensuring the integrity of the content.

5.2 Immutability Tests with Superuser (Root)

Subsequently, the same tests as before were performed, but with superuser (root) permissions, in order to validate whether immutability is maintained even with accounts with administrative privileges.

Case 4: Attempt to modify and delete as root

By repeatedly running the `rm`, `mv`, and `echo` commands on the immutable files, the system continued to display access denied messages, which showed that the protection was working correctly.

Case 5: Modifying permissions with `chmod`

An attempt was made to modify file attributes using the `chmod` permissions command. This was done by executing: `chmod monitoreo.62D2025... vbk` and then using `rm monitoreo.62D2025...vbk`.

Result: The file permissions were successfully modified, allowing for its subsequent deletion.

5.3 Problem solution investigation with superuser

Due to this identified vulnerability, an investigation was launched to find solutions that strengthen immutability, even for users with root privileges. This analysis focused on the following objectives:

1. Evaluate additional protection mechanisms in Linux systems (WORM, read-only mounting, extended attribute control).
2. Analyze the use of storage with native immutability (e.g., Object Storage with S3 Object Lock support).
3. Determine additional configurations in Veeam Hardened Repository to restrict access to the root user or apply `sudo` lockdown.

5.4 Test results

Test	Description	Expected Result	Actual Result	Conclusion
SSH connection	Verify that Veeam connects correctly to the Debian server	Stable and authenticated connection	Approved	Stable Connection
Repository detection	Confirm that Veeam detects the RAID storage	Correct display of volume/backup	Approved, after disk issues were resolved	RAID detected
File modification (standard user)	Attempting to edit an immutable file	Access denied	Access denied	Effective immutability
File deletion (standard user)	Attempting to delete an immutable file	Access denied	Access denied	Effective immutability
File movement	Attempting to move an immutable file	Access denied	Access denied	Effective immutability
Modification as root	Attempting to move an immutable file	Access denied	Access partially denied	Partially effective
Changing permissions with chmod	Alter immutable file permissions	Access denied	Allow change and deletion	Vulnerability detected

6. Error factors

During the process of configuring the immutable repository with Veeam Backup & Replication on Debian 11, multiple issues arose related to both the detection of storage disks and the validation of the file system and communication between Veeam and the server.

The main errors, their technical analysis, and the corrective actions applied are described below.

6.1 Disk detection error

When connecting the Debian server to Veeam, one of the disks in the RAID array was not recognized correctly. Although the RAID appeared to be in good condition in the BIOS and driver utilities, the operating system did not list all the volumes.

Using the commands: lsblk and parted -l

It was discovered that a disk was missing, preventing Veeam from identifying the full storage space. This was because the disk did not have a valid partition, so the procedure for manually adding the disk, previously explained, was followed. Initially, this process generated errors, but these were later corrected.

```

root@debianAlletra:/home/veeamrepo# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.36.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
The size of this disk is 36.4 TiB (40003191177216 bytes). DOS partition table format cannot be used on drives for volumes larger
than 2199023255040 bytes for 512-byte sectors. Use GUID partition table format (GPT).

Created a new DOS disklabel with disk identifier 0xaa6655a2.

Command (m for help):

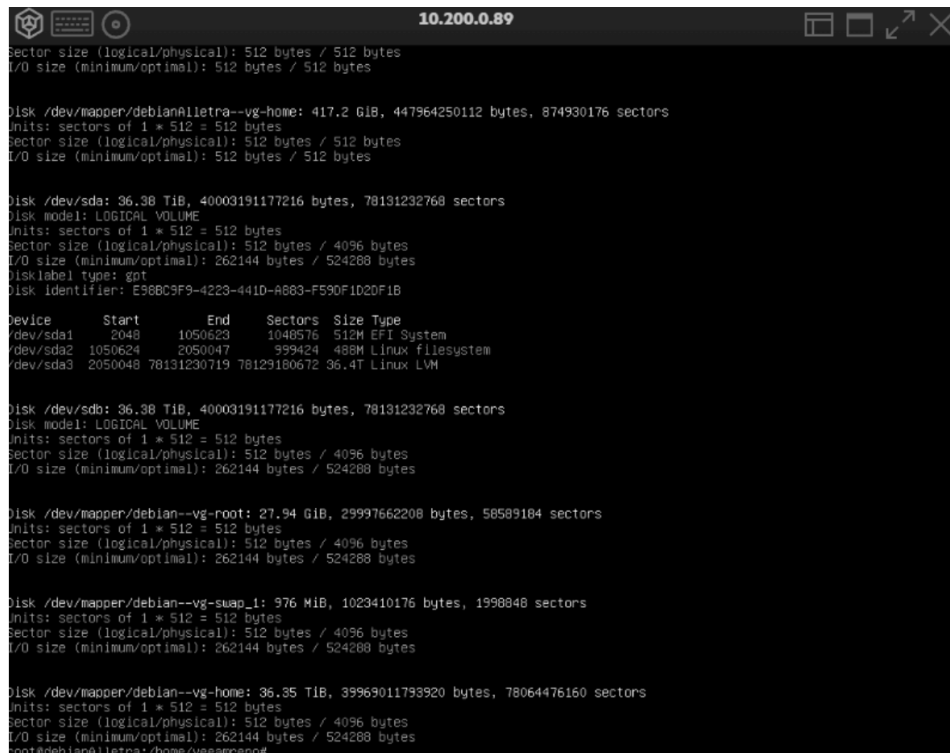
```

Figure 15. Error including disk

IN-LIFE OPERATIONS

VEEAM BACKUP CONFIGURATION FOR IMMUTABLE REPOSITORY

Once formatted and mounted correctly, Veeam recognized all of the storage.



```
10.200.0.89
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/debian-vg-home: 417.2 GiB, 447964250112 bytes, 874930176 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sda: 36.38 TiB, 40003191177216 bytes, 78131232768 sectors
Disk model: LOGICAL VOLUME
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 262144 bytes / 524288 bytes
Disklabel type: gpt
Disk identifier: E98BC9F9-4223-441D-A683-F590F1D20F1B

Device          Start      End          Sectors    Size Type
/dev/sda1        2048      1050623      1048576    512M EFI System
/dev/sda2       1050624    2050047      999424    488M Linux filesystem
/dev/sda3       2050048    78131230719  78129180672 36.4T Linux LVM

Disk /dev/sdb: 36.38 TiB, 40003191177216 bytes, 78131232768 sectors
Disk model: LOGICAL VOLUME
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 262144 bytes / 524288 bytes

Disk /dev/mapper/debian--vg-root: 27.94 GiB, 29997662208 bytes, 58589184 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 262144 bytes / 524288 bytes

Disk /dev/mapper/debian--vg-swap_1: 976 MiB, 1023410176 bytes, 1998848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 262144 bytes / 524288 bytes

Disk /dev/mapper/debian--vg-home: 36.35 TiB, 39969011793920 bytes, 78064476160 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 262144 bytes / 524288 bytes
root@debian:~#
```

Figure 16. List of available discs after correcting the error

6.2 Error with Veeam Transport service

During the initial connection from Veeam Backup & Replication to the Debian server, the process failed while attempting to install the Veeam Transport Service component.

The error indicated that the transport service could not be started or that the connection was interrupted during package transfer.

After reviewing the Veeam logs (C:\ProgramData\Veeam\Backup\Svc.VeeamTransport.log) and the Debian server log, the cause was determined to be a lack of stable SSH communication and the absence of certain packages required by Veeam (such as glibc and xfsprogs).

The fix involved reinstalling the SSH service and updating the system with the commands

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install openssh-server xfsprogs
```

After re-establishing the connection, Veeam was able to deploy the transport agent successfully.

6.3 Error en la conexión SSH

On several occasions, Veeam failed to establish an SSH connection with the Debian server, returning authentication or timeout errors. This problem was due to an incorrect initial configuration in the /etc/ssh/sshd_config file, where remote access was restricted.

The service was re-enabled, and root login was temporarily allowed (for controlled testing purposes only).

```
$ sudo systemctl enable --now ssh
$ sudo nano /etc/ssh/sshd_config
```

Subsequently, hardening measures were applied and the root shortcut was disabled again once the configuration was completed.

6.4 Error “File system behind the provided path is not XFS”

One of the most frequent mistakes when trying to add the immutable repository was:
Error: “File system behind the provided path is not XFS”

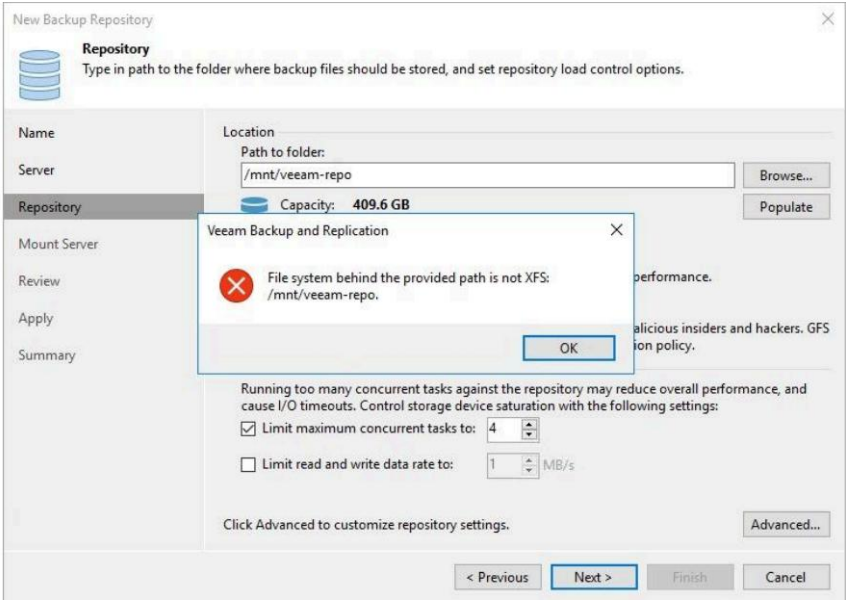


Figure 17. Error selecting folder

This message appeared when selecting the destination folder within the Veeam setup wizard. The error occurs because the immutability of Veeam Linux Hardened Repository requires that the file system of the designated directory be XFS and that it has reflink and d_type functionality enabled, properties necessary for Veeam to establish immutability attributes.

In this case, the solution consisted of formatting the disk with XFS, using the steps mentioned above in the installation process.

After the process was completed, Veeam recognized the path as valid and allowed immutability to be activated.

Name	Type	Host	Path	Capacity	Free	Used Space	Description
Alletra	Hardened	10.10.1.189	/home/veeamrepo	409.6 GB	388.7 GB	0 B	Created by WIN-H275MM1HMOI\Administrator at...

Figure 18. The folder was found in Veeam.

6.5 Immutability error with superuser (chmod)

Once the functional tests were completed, it was discovered that the root user could modify file permissions using the command: chmod.

This allowed for their subsequent deletion. This behavior is considered a critical vulnerability, as it defeats the purpose of file immutability if an attacker gains elevated privileges or if malware is executed with root privileges.

6.6 Summary of Errors Detected

Error	Causa	Impacto	Solución
Disk not detected by Debian	Missing partition in RAID volume.	Veeam did not recognize the storage space.	The partition was created and mounted manually.
Veeam Transport service failure	Missing dependencies and SSH errors during agent installation.	Veeam could not transfer data or create the repository.	Reinstalling openssh-server, base packages, and updating the system.
Connection failure	Incorrect sshd_config configuration.	Authentication from Veeam was not possible.	Adjusting SSH parameters and temporarily enabling root.
"File system behind the provided path is not XFS"	The disk was formatted without the reflink and crc options.	Veeam rejected the destination folder.	Formatting the file system with mkfs.
Immutability violated with chmod	Superuser can modify permissions in XFS.	Risk of deletion of backups.	Documented and referred for further investigation. Documented and referred for further investigation.

6.7 Safety aspects

- SSH access was configured using only key authentication, and direct root access was disabled (PermitRootLogin no).
- Read-only permissions were applied to the immutable folder.
- The encrypted connection between the Debian server and the Veeam server was verified.
- Implementing log monitoring (/var/log/auth.log) and change auditing was recommended.
- To prevent future modifications with chmod, it was suggested to implement a temporary read-only mount or migrate to storage with native immutability (e.g., object storage compatible with S3 Object Lock).

Control de versión

Versión	Fecha	Author	Descripción
1.0	02/10/2025	Pablo César G.	Network requirements and topology
1.1	09/10/2025	Pablo César G.	Installation and configuration of the service
1.2	16/10/2025	Pablo César G.	Errors encountered during installation